

ABSTRACT

A way for computer programs to shorten and summarize longer texts Forwarding the intended message is defined as Automatic Text Summary and a General Problems in Machine Learning and Natural Language Processing (NLP) .It has become an important area of research in natural language processing as it enables people to digest text more quickly and efficiently .Text Summarizing is the process of making longer, shorter, and sequential summaries including text documents and presentation of the main points of the text. identify text, interpretation and summary, some summary analysis is produced one of the main difficulties in the process of summarizing the text. There are three main steps to compile documents. This is a topic definition, description and summary generation Identifying topics: The most prominent information in the text is identified. There are several ways to determine the subject, i.e., Position, Cue Sentence, word frequency. The method is based on the most common sentence position useful technique to determine the topic. There are two main methods of scripting: extract and complete. Extracting content involves selecting the most important phrases or sentences from the original text and putting them in context. Summarization on the other hand, involves creating new sentences that capture the meaning of the original text, often using language construction techniques. While abstract content is generally easier to use, abstract content has the potential to create more complex and readable content. It has many applications such as short text, news gathering, automatic data collection for search engines and research.

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Chapter 1

INTRODUCTION

1.1. Problem Statement and Objective

To explore and study various text summarization techniques that is available and implement them. To be able to take inputs from semantic web (from the user) and get the summarized text from it. Generate relationships between various entities in the summarized text. Generate the graph based on the relationships.

Extracting the information from semantic web and apply text summarization for the purpose of better understanding. Semantic web is a proposed development of the World Wide Web in which data in web pages is structured and tagged in such a way that it can be read directly by computers. Examples include Best Buy, BBC World Cup site, Google, Facebook and Flip-board.

1.2. Text Summarization

Text summarization is the process of creating short, smart and most importantly points from longer texts. The main idea behind the automatic collection of content is to be able to find a small piece of the most important information from the entire collection and present it in a way that people can read. As information online increases, automatic content authoring can be beneficial because more useful information can be read in a short time. It may be tempting to use a table of contents for all texts to extract useful information and reduce reading time. However, in its current form, NLP content only has a limited number of applications. Writing a text is very effective if the text contains many facts and can be used to filter important information from it. An NLP model can collect long information and represent it in small sentences. News, fiction, and posts all fall under this category. However, for text where each sentence is based on the previous sentence, the text does not work well. Research papers, medical articles are examples of good literature for which abstracts would not be complete. Finally, if we consider the example of summarizing novels, the summarization method is successful. However, it may miss the style and tone of the text the author is trying to convey . Therefore, writing can only be used in a few applications. This is how the extract summary method works. It uses the text, sorts all sentences according to the clarity and accuracy of the text, and presents you with the most important sentences.

1.3. History of text summarization

Early in the 1950s, when scholars first started looking at ways to automatically create summaries of news stories, the history of text summarizing was established. At the University of Pennsylvania, the News Text Analyzer (NTA), the first text summarization system, was created in 1958. To extract crucial phrases from news items and provide summaries, the NTA adopted a rule-based methodology. However, early text summarization systems were constrained by a lack of computational power and the comparatively rudimentary natural language processing approaches that were available at the time.

The study of text summarization accelerated in the 1990s with the introduction of more potent computers and the creation of more sophisticated natural language processing methods. As a result, numerous fresh summarizing methods were created, including statistical, machine learning, and graph-based approaches. These methods set the path for the creation of contemporary text summarizing systems by making it possible to summarize text documents in a more accurate and effective manner. Text summarizing is a current technology that is still under active study and development. It is utilized in a wide range of applications today, including news media, academic research, and corporate intelligence.

1.4. NLP

The subject of computer science and linguistics known as "natural language processing" (NLP) is concerned with how computers and human language interact. It entails creating computer models and methods that allow machines to comprehend, decode, and produce natural language. Language translation, sentiment analysis, chatbots, and automatic summarization are just a few of the useful applications of NLP. The foundation of NLP is a blend of computer methods and language expertise. chatbots, and automatic summarization are just a few of the useful applications of NLP. The foundation of NLP is a blend of computer methods and language expertise. It involves parsing and evaluating language at different complexity levels, including syntax, semantics, and pragmatics. The complexity and variety of human language is one of the main obstacles for NLP. This may be caused by elements including syntax, word sense, context, and culture. Numerous methods, including machine learning, deep learning, and statistical analysis, are employed by NLP researchers. Additionally, they depend on linguistic tools like lexicons, grammars, and ontologies. Despite these difficulties, NLP has made significant strides lately, and its applications are spreading rapidly.

1.5. Extractive Summarization

The practice of condensing a larger text while maintaining its main ideas is known as text summarizing. Abstractive and extractive summarization are the two basic methods used to summarize texts. Extractive summarization entails choosing and combining the most crucial phrases or sentences from the original text to create a summary. In contrast to abstractive summarization, this method does not require the model to produce new text, making it easier to use. Instead, it concentrates on extracting the key details from the original text and summarizing them. Extractive summarization frequently uses statistical and machine learning methods including graph-based algorithms, ranking, and clustering. Extractive summarization can result in high-quality summaries, but it might miss some of the subtleties of the original text or fail to give the information in a more compact manner. [1]

1.6. Abstractive Summarization

Abstractive summarization entails creating a summary that is not merely a duplicate of the original text's words but rather a new text that concisely expresses the important ideas. For this, the model must comprehend the text completely and be able to come up with new sentences that express the same ideas. As the model must produce coherent and grammatically sound sentences that do not appear in the source text, abstractive summarization is more difficult than extractive summarization.

1.7 Semantic Web

Both Semantic Web technologies and Natural Language Processing (NLP) are Semantic Technologies, yet they play distinct and complementary roles in data management. In reality, businesses may mix structured and unstructured data in ways that are just impractical with conventional techniques thanks to the combination of NLP and Semantic Web technology.

A group of technologies known as the Semantic Web are used to represent, store, and query data. Although textual data can be stored using these technologies, such as text in a Word document or PDF file, they are typically used to store smaller bits of data. Consequently, the Semantic Web also contains numbers, dates, figures, and other data in addition to text, whereas Semantic Search primarily focuses on textual information. For instance, unlike on your computer or on a single web page, you wouldn't store the full page as a single value in the Semantic Web if you wanted to represent a collection of facts about the White House using RDF (the data model for the Semantic Web). The Address, Date Constructed, Designer, Current Resident, etc. would be kept as precise factual information instead.

This flexibility is meant to make it easier for us to find solutions to challenging inquiries such, "Which President who lived in the White House had at least one child who did not live with him in the White House?" Such complex queries involving various forms of data from various sources in various formats are well- suited for the Semantic Web to resolve.

1.8 Comparison between extractive and abstractive summarization: There are two commonly used methods for summarizing text documents: extractive and abstractive. In order to construct an extractive summary, significant sentences or phrases from the original material are chosen and concatenated. This method, which is rather simple, yields summaries that nearly mirror the original text. The objective of extractive summarizing, which is frequently employed in news items, is to extract the most crucial information and convey it succinctly. Extractive summarization, however, is limited in its ability to handle complicated or multifaceted manuscripts, where crucial information may be scattered among a number of lines or paragraphs.

Advanced natural language processing methods, such as text production and semantic comprehension, are needed for this methodology. Because it can pull information from various parts of the document and rephrase it, abstractive summarization can produce summaries that are more succinct and coherent than extractive summarization. In contrast, abstractive summarizing is more difficult than extractive summarization since it calls for in-depth knowledge of the original text as well as the capacity to produce grammatically accurate and semantically meaningful phrases. Despite these difficulties, personal assistants, chatbots, and document summarization applications all use abstractive summarization more and more frequently.

1.9 Advantages of text summarization

There are several benefits to text summary in a variety of industries, from the media to academic study. The fact that text summaries can save time and resources is a big benefit. It might take a lot of time to summarize large documents, such news stories or research papers. However, the procedure can be accelerated with the aid of automated text summarization methods, enabling more effective use of time and resources. By giving readers, a brief rundown of the key ideas in the text, summarization can also help readers avoid information overload by enabling them to decide right away whether the document is pertinent to their interests.

Text summary also makes information more accessible and increases understanding. Long texts, especially ones with complicated subject matter, aren't always worth the time or effort to read. Such people can access important information through summarization without having to read the

entire document. Additionally, summaries are typically written in plainer language than the original text, which makes them simpler to comprehend. The ability to swiftly understand the key ideas of a complicated subject by reading a summary makes text summarizing extremely helpful in teaching and research. Overall, there are many advantages to text summarization, such as time and resource savings, a decrease in information overload, and improved accessibility and comprehension.

1.10 Advantages of representing text in graph

In applications involving machine learning and natural language processing, representing text as a graph provides various benefits. Its capability to handle and analyze vast volumes of text data quickly is a key benefit. It is simpler to see patterns and relationships within the text when it is transformed into a graph, where nodes stand in for words or phrases and edges signify links between them. In applications like sentiment analysis, where the objective is to recognize and assess the emotional tone of a significant amount of text input, this might be quite helpful. The ability to create more complex natural language processing models is another benefit of encoding text as a graph. To recognize intricate patterns and correlations in text data, graph-based models can be built. This enables more precise forecasting and analysis. In order to provide more accurate summaries or to categorize documents based on their content, graph-based models, for instance, can be used to identify essential concepts and entities inside a text.

A third benefit of portraying text as a graph is that it may make information retrieval and search more effective. Instead of depending on keyword matching or other less complex techniques, graph-based search engines may swiftly find the most pertinent items based on their content. This can be especially helpful in applications like information retrieval or question-and-answer systems, where the objective is to discover the most pertinent material within a sizable text database.

Overall, expressing text as a graph has various benefits, such as effective text data processing and analysis, the development of more complex models for natural language processing, and more effective information search and retrieval.

Chapter 2

LITERATURE REVIEW

A literature survey is a methodical and thorough study of a variety of published literature and other information sources, including theses, to find as many pertinent articles on a specific topic as possible. Consider these articles before moving on with the project.

Extractive text summarization using sentence ranking. [1]

Input: Data as input as text.

Output: Create text content correctly, shorter than original. The extracted content is converted to audio format.

Dataset: The current dataset.

Method: In summarizing, the summative processes the text using the input and tokenizes the text to retrieve the information. To filter the text, stop removing the message. Finally, each icon is given a speech bubble.

Step 1: After adding some of the speech tags to coins or dots, each person's weight is assigned to the coin. The time weight is calculated as: $W_t = (1)$

Step 2: Now consider the maximum weight of the coin after finding the maximum weight.

The weight of the data is calculated as: $W_{tf} = (2)$

Step 3: In this step, combine and tally the frequency of the corresponding words in the sentence.

Step 4: Finally, the extractor will extract the top sentence from the file and convert the extracted sentence to audio (mp3) format.

Implementation: The program has been tested with five files totaling twenty sentences. The creation of sentences with a level greater than 8 begins with content-

based outputs. Audio content is created from text. The written material is implemented using NLTK and Python 3.6.

To get expression markers, the data provided as input is first tokenized. Stop words are eliminated from the text after symbolization. Keywords are regarded as significant content. Since we add a portion of the tag to each keyword, content is considered as input.

Now that the frequency of the keyword has been divided, we can determine the weight of a word based on its frequency.

The high weighted frequency sentences will then be retrieved by the summarizer, and they will then be transformed into audio.

1.1 EXTRACTIVE TEXT SUMMARIZATION

A. Term Frequency-Inverse Document Frequency Method

A word's importance in a particular document is shown by its term frequency (TF) and inverse document frequency (IDF), two numerical statistics. TF is the number of times a term appears in the document, and IDF is a metric that increases the weight of terms that occur infrequently and decreases the weight of phrases that appear frequently in the collection. Following that, each sentence is given a score based on its product, and those with a high score are included in the summary. This approach has a flaw in that lengthier sentences occasionally receive high marks.

B. Cluster Based Method

A form of recommendation system methodology known as the cluster-based method divides individuals or things into clusters based on how similar their tastes, behaviors, or other traits are to one another. The technique identifies groupings or clusters of similar users or items by analyzing data from several people or items. Using distance measures like Euclidean distance or cosine similarity, the cluster-based method often determines how similar users or items are. After the similarities have been determined, users or items can be grouped into clusters using clustering algorithms like k-means, hierarchical clustering, or density-based clustering. The cluster-based technique can produce tailored suggestions for users based on their preferences and those of other users or products in the clusters once they have been identified. The cluster-based technique can produce tailored recommendations for users based on the preferences of other users or objects within the same cluster once the clusters have been detected. For instance, if a user is part of a group of users with comparable preferences, the system may be able to suggest products that are well-liked or well- rated among that group

.

C. Text Summarization with Neural Network

A Neural Network is a processing system that attempts to mimic the learning process of the human brain by modelling it after that organ. A neural network is a network of connected artificial neurons that processes data using a numerical model of computing. When summarizing text, the strategy getting the neural systems ready to recognize the kinds of sentences that should be included in the summary. Each sentence in the test paragraph is examined to see whether it should be included in the summary before being used to train the neural network. Summary sentences are correctly classified by neural networks, but they struggle with lengthy training periods.

D. Text Summarization with Fuzzy Logic

A method for automatically producing text summaries that are more readable and human-like is text summarization with fuzzy logic. Fuzzy logic is a mathematical method for dealing with uncertainty and imprecision, and it can be used to convey the meaning of text in a more subtle and adaptable way than conventional binary logic.

In text summarizing with fuzzy logic, the text is initially examined using natural language processing methods to determine the key phrases and ideas in the document. Then, based on variables like the frequency of occurrence, the context in which it appears, and the semantic relationships between various concepts, fuzzy logic is used to assign a level of relevance or importance to each sentence or concept.

The text summarizing algorithm chooses the most significant sentences and combines them into a summary to capture the main ideas of the original content after each sentence or notion has been given a level of relevance. Additional information, such as key terms or ideas that were not expressly stated in the original text, may also be included in the summary.

E. Graph based Method

The relationships between people, items, and other entities are represented using graph structures in the graph-based method, a form of recommendation system technology. In this approach, the graph nodes stand in for entities like users, items, and attributes, while the edges between nodes signify connections.

When there is a vast amount of data to be analyzed and when people have complicated preferences and interests, the graph-based approach performs 1) Tree Based Method: Tree based method uses dependency tree to represent text document. Source text is first represented as dependency trees then these trees are exceptionally well. By recognizing related objects and recommending them based on their relationships in the graph, it can also be used to produce suggestions for new and previously undiscovered items. In the graph, each user and item is represented as a node, and the edges signify how users and items interact with one another. The edges may be binary if the interaction took place or they may be weighted according to the strength of the interaction. A graph is created using the user-item interaction data in the graph-based method.

F. Query based summarization

The appropriate amount of the necessary information is provided through query- based text summarization in accordance with the search query entered by the user. As a result, the user doesn't have to spend a lot of time looking up the information they need. The sentences in a known document are graded using criteria like phrase frequency counts in this summarizing method. 5 The sentences with the most query expressions receive better scores than the ones with the least. The output summary is then combined with the sentences with the highest scores. Results from query-based text summarization are precise. A query that only uses a few small phrases may result in the loss of significant information.

1.2 ABSTRACTIVE TEXT SUMMARIZATION METHOD

A. Structured Based Approach:

1) Tree Based Method: Dependency trees are used in tree-based methods to represent text documents. Initial representations of source text take the form of dependency trees, which are subsequently combined into a single tree before being transformed into what is known as a fused sentence. Tree linearization is the process of reducing a dependency tree to a string of words. The effectiveness of tree-based summarization depends on the parser selected and the degree of word dependency that is maintained.

2) Template Based Method: In template-based techniques, the document is represented by a template. To identify text content that has been mapped into template space, text is matched in opposition to patterns and rules. Different template-based systems have different linguistic coverage, syntactic familiarity, and filling-out procedures. The content of the summary is indicated by text that fits inside the form. The summary produced using the template-based method is quite well- coordinated. Semantic analysis is one of the key issues with template-based methods since templates are fairly specific in the sense that they accept only highly relevant content and demand depth.

3) Ontology Based Method: An ontology is a formal name and characterization of the various entity types connected to a certain topic that serves as a knowledge base. A knowledge base is employed in this strategy to enhance the effectiveness of the summarization. The majority of internet documents belong to a certain domain with a small vocabulary, which the ontology can better express. We can enhance the semantic representation of information content and query expansion using ontology attributes.

4) Lead and Body Phrase Method: This approach is phrase-based. Main sentences, or those that

are informative in context and have a good length, are rephrased using the lead and body phrase method by inserting and substituting phrases. This approach works well for lead sentence adjustments that are semantically suitable. The performance degradation caused by parsing and the lack of a universal model for summarizing is one of the main shortcomings of lead and body phrases.

5) Rule Based Method: The information extraction rules that the user has expressly stated are used to pick content in the rule-based method. Last but not least, summary phrases are created using linguistic patterns. This method's strength is that it yields summaries with more information in them. The primary flaw is that each rule and pattern must be manually developed, which takes a lot of time and effort.

B. SEMANTIC BASED APPROACH

1) Multimodal semantic model: When information is presented in various modalities, such as text, photos, videos, and audio, a multimodal semantic model is a specific kind of machine learning model that is created to comprehend and express the meaning of the information. To produce a more thorough comprehension of the data, the model develops the ability to recognize and examine the connections between various modalities. Natural language processing, image and video classification, speech recognition, and recommendation systems are just a few applications for multimodal semantic models. The characteristics and connections between various modalities are often learned by these models using deep learning approaches like convolutional neural networks (CNNs) and recurrent neural networks (RNNs).

2) Information item-based method: In this method, a recommendation system methodology known as the information item-based method creates customized recommendations for users based on their preferences and prior interactions with things. The technique finds other items that share similar features by examining the attributes of the objects the user has interacted with. Items are characterized using a variety of properties, including genre, author, rating, and length, in the information item-based method. The method then determines how similar the qualities of various things are by using algorithms like cosine similarity or Pearson correlation coefficient.

The recommendation system can produce a list of products that are most comparable to the objects that the user has already engaged with after calculating the similarities between items. The user can then see this list as tailored recommendations after it has been graded based on how comparable the items are.

3) Semantic Graph Based Method: A form of recommendation system technique known as semantic graph-based method makes use of graph structures to express the connections between users, objects, and other things. In this approach, the graph nodes stand in for entities like users, items, and attributes, while the edges between nodes signify connections between those entities. The semantic graph-based approach identifies the qualities of people and items and builds a semantic representation of their preferences and traits using machine learning and natural language processing techniques. The graph structure is then built using this semantic description.

The semantic graph-based approach determines the relevance and significance of various nodes and edges in the graph using methods like personalized PageRank or random walks with restarts once the graph structure has been established. This allows the method to generate personalized recommendations for users, based on their interactions with items and their relationships with other users.

1.3 Text summarization using text frequency ranking sentence prediction:

Input: Amazon product review dataset is used to find the summary of a particular product. The raw input data collected from different sources are pre-processed through various libraries in NLP that is used for string or text processing.

Dataset: Amazon product review dataset obtained from Kaggle

Methodologies: A combination of supervised and unsupervised learning methods is employed in the text frequency ranking sentence prediction system. Uses a combination of the supervised algorithm Seq2Seq and the unsupervised algorithm TF-IDF-TR (Text frequency - Inverse document frequency - Text rank). Aims to profit from abstract and extractive summarization.

Implementation: The dataset is pre-processed (stop words, lemmatization, elimination of null values, etc.) and fed as input to the TF-IDF-TR. The extractive summarization result is then fed as input to the abstractive summarization-Seq2Seq model. Since extractive summarization cannot handle large documents, the size of the original document is decreased after extractive summarization, which is advantageous to abstractive summarization. Using the ROGUE score, this output is compared to other Text summarizing techniques that are currently in use.

1.4 Speech Document Summarization using Neural Network:

DATASET: A recorded speech which is converted into a text document.

METHODOLOGY: A model that combines extractive and abstract modelling techniques. There are two subparts in the extractive summarization section. Hierarchical sentence representation is

one example. We eliminated the stop words from the manuscript during the pre-processing of the data. Each sentence is then broken apart, and the sentences are tokenized. Do the word embedding after that. Word embedding is a sort of word representation that enables the depiction of words with related meanings. Here, I used a pre-trained embedding and performed a 128- dimensional word embedding.

We employ a convolutional model to perform sentence representation, where the convolution sentence encoder encodes each sentence in the text. To compute the representation of the entire document, we incorporate a bidirectional LSTM-RNN (Long Short-Term Memory Recurrent Neural Network). Before generating a document summary, the entire document is read to fully comprehend its content. The Bi-LSTM-RNN plays a crucial role in understanding the document as it examines both past and future sentences, making it more effective than a general LSTM-RNN in encoding textual data.

Once the document is encoded, the next step is **sentence selection**. In this phase, important phrases are extracted from the sentence representations. To enhance the accuracy of selection, an additional LSTM-RNN is introduced. This model considers both the prior output and the current input while extracting key sentences, ensuring coherence in the summary.

The LSTM model operates using three essential gates: **input, output, and forget gates**. These gates regulate the flow of information, allowing the model to retain essential details while discarding redundant data. The LSTM mechanism is particularly effective in identifying crucial passages from the source text, ensuring that the generated summary maintains its relevance and completeness. Additionally, further improvements can be made by refining sentence scoring techniques and integrating attention mechanisms to enhance the summarization process.

SI NO	PAPER TITLE	AUTHORS	DESCRIPTION
1	EXTRACTIVE TEXT SUMMARIZATION USING SENTENCE RANKING	J.N. Madhuri; Ganesh; Kumar.	Summarizer will extract the high weighted frequency sentences and the extracted sentences are converted into audio form
2	MULTI-DOCUMENTTEXT SUMMARIZATION	Amol Tandel; Brijesh Modi; Priyasha Gupta; Shreya Wagle; Sujata Khedkar.	Cross document relation using voting technique - it is an efficient method to identify relations between sentences.
3	SPEECH DOCUMENT SUMMARIZATION USING NEURAL NETWORK	Kirusiha Balasundarm ;C.R.J.Amal raj.	In this paper a model which combines both extractive and abstractive techniques have been used.

4	EXTRACTIVE TEXT-IMAGE SUMMARIZATION USING MULTI- MODAL RNN	Jing Qiang Chen; Hai Zhuge.	The sentences and documents are encoded by a hierarchical bi-directional RNN model. The vector representation of the images is extracted using VGGNet.
5	TEXT SUMMARIZATION IN INDIAN LANGUAGES: A CRITICAL REVIEW	Nomi Baruah; Shikhar ; Kr. Sarma; Surajit Borkotoy.	This paper reviewed the existing text summarization approaches done on a few Indian languages.
6	IMPROVING THE QUALITY OF TEXT SUMMARIZATION USING PRONOUN REPLACEMENT TECHNIQUE	Siddha Ling Urolagin & Likitha Satish.	The method used for automatic summarization in the paper is based on identification of keywords and frequency of keywords Methodology
7	A SURVEY ON NLP-BASED TEXT SUMMARIZATION FOR SUMMARIZING PRODUCT REVIEWS	Ravali borugu and G Ramesh.	An extractive based Text Summarization technique that makes use of Genetic Algorithms.

8	AUTOMATIC TEXT SUMMARIZATION USING INTERNAL AND EXTERNAL INFORMATION	Kamal Sarkar.	TF IDF method has been used for measuring internal content of each sentence in a document.
9	ANALYTICAL STUDY OF TEXT SUMMARIZATION TECHNIQUES	Pooja Raundale; Himanshu Shekhar.	TF-IDF, Text rank, Seq2Seq with attention model, Pointer Generator are the techniques used.
10	NATURAL LANGUAGE PROCESSING-BASED ABSTRACTIVE TEXT SUMMARIZATION OF REVIEWS	Janice Shah; Manasvi Sagathiya; Kareena Redij; Varsha Hole.	They use Seq2Seq with an attention model that focuses on features of text for summarization. As an output the system generates the summarized version of the reviews.
11	AUTOMATIC VIDEO SUMMARIZATION WITH TIMESTAMPS USING NLP	Ahmed Emad; Fady Bassel; Mark Refaat; Mohamed Abdelhamed ; Nada Shorim; Ashraf AbdelRaouf.	CNN, LSTM, sentence segmentation, lemmatization and stemming and abstractive summarization are the techniques used.

Chapter 3

METHODOLOGY

3.1 Proposed System Architecture

Following figure depicts the system architecture. Model has three modules the first module is used to produce the summary of the given input text. Output of one module is given as input for another module.

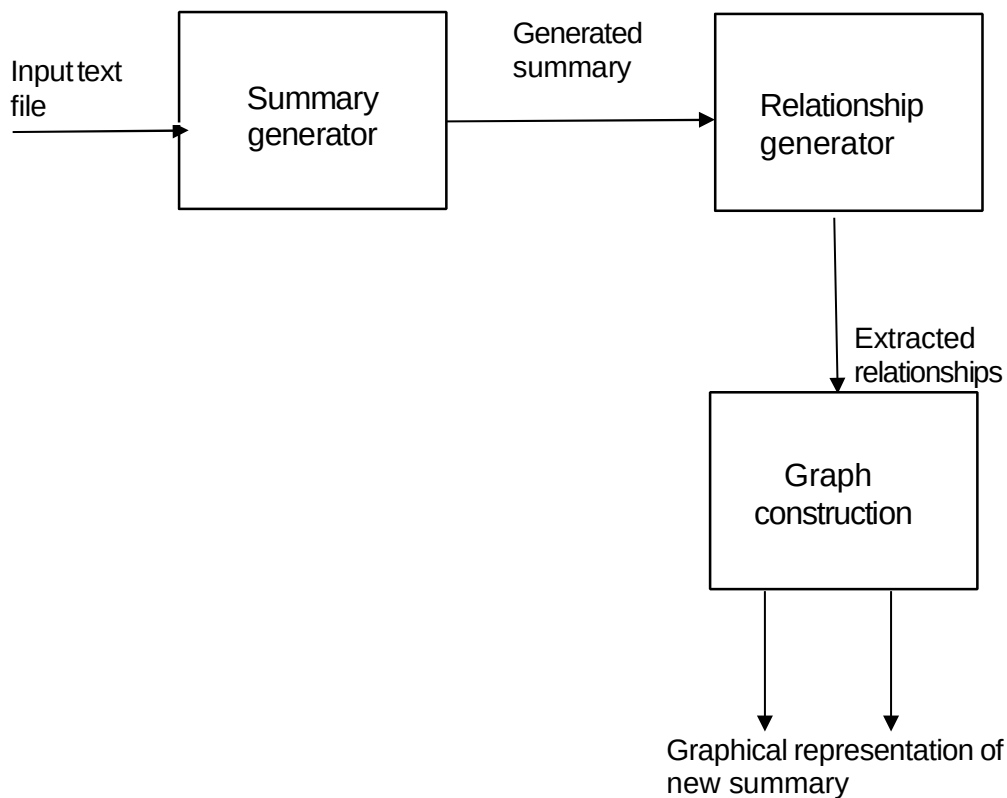


Figure 3.1 Proposed system architecture

3.2 Explanation of the Architecture

The project operates in multiple phases, beginning with the **input phase**, where the user provides a text to be summarized. The input is processed by the **summary generator**, which creates a concise version of the text. Based on our research, we have identified two main text summarization techniques: **extractive** and **abstractive**. Extractive summarization is chosen for this project as it is more effective when dealing with structured information. On the other hand, abstractive summarization is more suitable for conversational text, but it is not the preferred approach in this case.

Once the summary is generated, it is passed to the **relationship generator**, which identifies key

relationships and keywords from the summarized text. These extracted relationships are then supplied to the **graph generator**, which creates a visual representation of the summarized content. This graph provides a clear and structured overview of the input text, making it easier to interpret.

Additionally, **sub-graphs** are generated based on the importance of various entities present in the text. These sub-graphs are later merged to form a **comprehensive knowledge graph**, offering a more detailed and structured representation of the relationships within the summarized text. This approach enhances the overall understanding and visualization of the input data.

To further enhance the effectiveness of the summarization and visualization process, additional techniques can be incorporated. **Natural Language Processing (NLP) techniques** such as named entity recognition (NER) and part-of-speech tagging can help in better understanding the context of the text. These techniques allow the system to accurately extract important entities, such as names, locations, and dates, which contribute to the overall structure of the knowledge graph.

Moreover, **sentence ranking algorithms** can be employed to prioritize more relevant sentences in the extractive summarization process. Methods such as **TextRank, LexRank, and TF-IDF** can be utilized to ensure that only the most informative sentences are selected. By integrating such algorithms, the quality of the summary improves, making it more meaningful and concise.

To refine the **relationship extraction process**, machine learning models such as **BERT and GPT-based transformers** can be leveraged to detect complex relationships between entities. These models enable a deeper semantic understanding of the text, improving the accuracy of relationship identification and ensuring that the generated graph is more informative.

Furthermore, integrating **interactive visualization tools** will enhance the usability of the generated knowledge graph. By allowing users to interact with the graph, zoom into specific entities, and explore different relationships, the system becomes more intuitive and insightful. Future improvements could also involve **real-time summarization**, where text streams (such as news articles or live transcripts) are processed dynamically to generate evolving knowledge graphs.

By combining these advanced techniques, the project can achieve a more robust, scalable, and intelligent summarization system that not only condenses information efficiently but also provides meaningful insights through structured visualizations.

Chapter 4

IMPLEMENTATION

4.1 STUDY AND ANALYSIS OF VARIOUS TEXT SUMMARIZATION TECHNIQUES

4.1.1 PEGASUS (Pre-training with Extracted Gap- sentences for Abstractive Summarization)

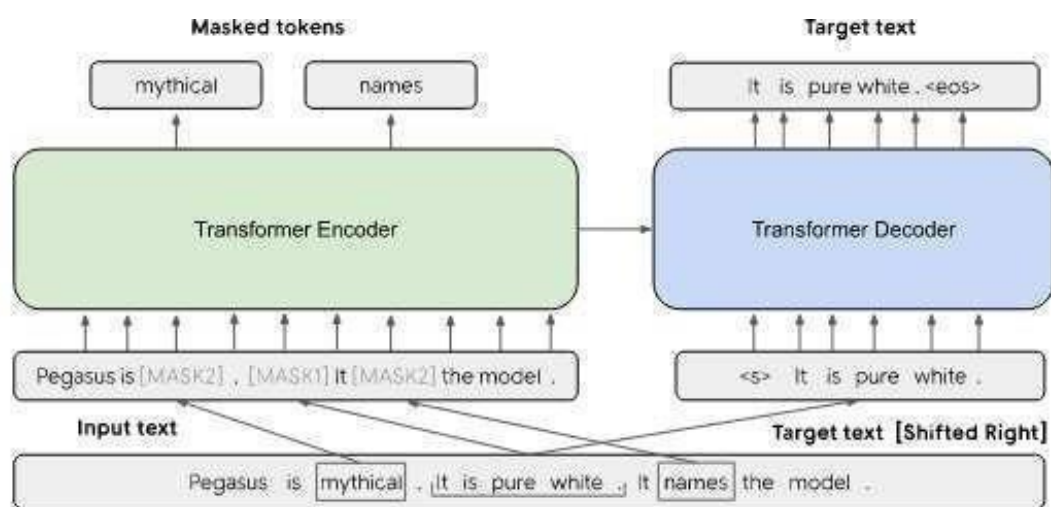


Figure 4.1.1 Pegasus model

PEGASUS uses an encoder/decoder model for sequence-by-sequence learning. At the same time models, encoders first consider the full text of the input text, encodes the input text into what is called a context vector. This is basically a number representation of the input text, this numerical representation is given to the decoder. Its task is to decode the context vector and create a summary. Where PEGASUS differs from previous state-of-the-art models is pre-training. You can use pre-trained models and fine-tune the model on real data of much less amount. In fact, the evaluation results are for different data sets with only 1000 training data, this model achieved comparable results to previous SOTA models. This has important practical implications because many of us do not have the resources to collect tens of thousands of document-summary pairs.

4.1.2 TF – IDF (Term frequency Inverse document frequency)

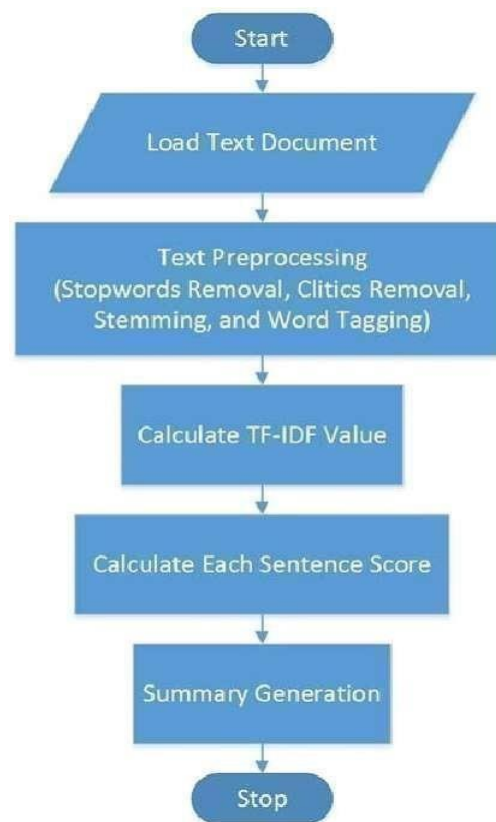


Figure 4.1.2 Term frequency Inverse document frequency

Tokenize the sentences: We'll tokenize the sentences here instead of words, and we'll give weight to these sentences.

Create the Frequency matrix of the words in each sentence: We calculate the frequency of words in each sentence.

Calculate Term Frequency and generate a matrix: We'll find the Term Frequency for each word in a paragraph. Now, remember the definition of TF,

$$TF(f) = (\text{Number of times the letter } f \text{ appears in a document}) / (\text{Total number of letters in the document})$$

Creating a table for documents per words: This again a simple table which helps in calculating IDF matrix.

Calculate IDF and generate a matrix: We'll find the IDF for each word in a paragraph. Now, remember the definition of IDF,

$$IDF(f) = \log_e (\text{Total number of documents} / \text{Number of documents with letter } f \text{ in it})$$

Calculate TF-IDF and produce a matrix: Now we have both the matrix and the next step is simple. TF-IDF algorithm is made of two algorithms multiplied together. In simple words, we are multiplying the values from both the matrix and generating new matrix.

Score the sentences: Scoring a sentence is different with different algorithms. We are using TF-IDF score of words in a sentence to give weight to the paragraph.

Find the threshold: Similar to any summarization algorithm, there can be multiple ways to Calculate threshold value. We are calculating average sentence score. Then finally produce the summary. [9]

4.1.3 LSA (Latent Semantic Analysis)

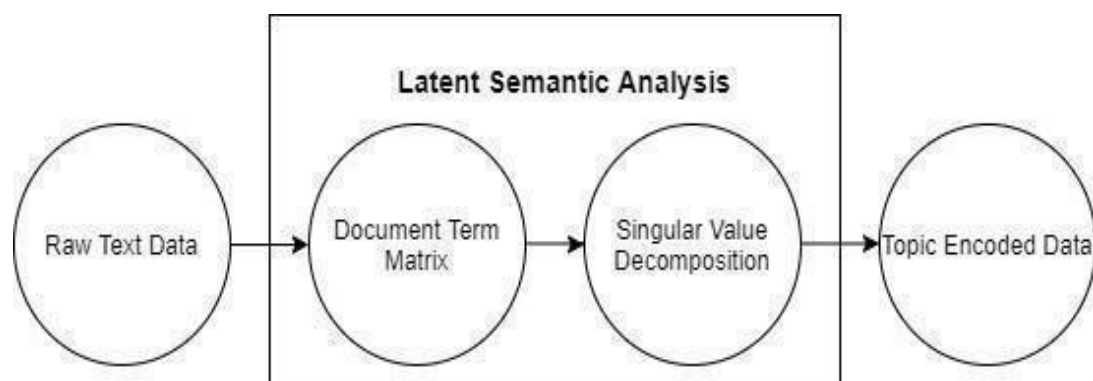


Figure 4.1.3 Latent Semantic Analysis Model

One of the statistical NLP techniques is Latent semantic analysis. It is an approach to determining associations between words in a document. LSA is engaged with problems of the following types: Example: mobile phone, phone, cell phone, phone are similar, but the following query only documents containing "cell phone rang" and "cell phone" are retrieved and documents containing mobile, phone, and phone are not searched.

LSA - Assumptions:

1. Words used in the same context are similar to each other.
2. Hidden semantic data structures are not clear due to the word ambiguity selected.

Singular Value Decomposition: Singular value decomposition is a statistical method.

It is used to find latent (hidden) semantic structures of words scattered throughout a document.

4.1.4 Text rank using cosine similarity

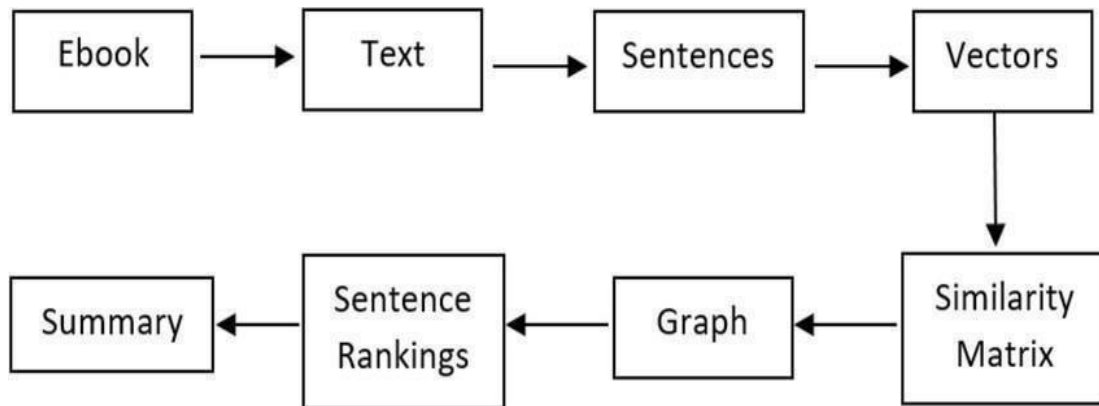


Figure 4.1.4 Text rank sequence

Text Rank is a graph-based ranking algorithm for natural language processing. This is an extractive, unsupervised approach to text summarization using a similarity matrix and use page rank to identify similarities between texts and rank text graphically ranking algorithm. A similarity matrix is used to graphically display text whose vertices are: text and edges are links between weighted text.

Steps

1. Covert text to sentences: Converting a single text to a list of sentences.
2. Pre-process text: Clean the sentences by removing unnecessary words, stop words, punctuations, etc.
3. Extract word vector embedding: Word embedding represents the vectors of words which are the mathematical form of a word. It represents words as real-valued vectors in a predefined vector space.
4. Generate sentence vector embedding: Using the word vector embedding, generate sentence vector embedding by averaging the sum of word vectors.
5. Create similarity matrix: Similarity matrix represents the similarity between each sentence with every other sentences.
6. Determine sentence rank: It is determined graphically by using page rank algorithm.
7. Generate Summary: Generate a summary by extracting the sentences having higher Ranks [10]

4.1.5 Lex rank

LexRank is an unsupervised graph-based approach for extractive text summarization, closely related to TextRank but with key differences in how it measures sentence similarity. Unlike TextRank, which primarily relies on Jaccard similarity, LexRank employs Cosine similarity, determining the angular distance between sentence vectors rather than simply measuring word overlap. This approach provides a more accurate representation of semantic closeness between sentences, ensuring that only meaningful relationships are established. The fundamental idea behind LexRank is to construct a weighted, undirected graph where each node represents a sentence, and the edges between them are weighted based on their similarity scores. A threshold-based filtering technique is applied to ensure only strong sentence relationships are included, preventing weak connections from distorting the ranking process. Once the graph is built, eigenvector centrality is used to determine the most important sentences, ensuring the summary remains informative while minimizing redundancy.

Although LexRank is widely used in extractive text summarization, it has limitations. Since it relies on TF-IDF and Cosine similarity, it may struggle to capture contextual nuances and deep semantic relationships between words. Its effectiveness also depends on the threshold parameter, which, if set incorrectly, could either exclude important sentences or include irrelevant ones. LexRank builds upon TextRank, improving sentence selection by incorporating lexical centrality, making it more effective at reducing redundancy.

4.2 ALGORITHM

Input: Text that needs to be summarized Output: Summarized text in the form of a graph

Step 1: Text given by the user which needs to be summarized is given to the summary generator.

Step 2: The output of the previous step is summarized text which is served as an input to the relationship generator.

Step 3: The relationship generator gives you various relations that are present in the summary which then becomes input to the graph generator.

Step 4: Using the relationships generated in the previous step graph is constructed. Step 5: Some important entities from the graph are picked.

Step 6: For the important entities separate sub-graphs are generated.

4.3 DATASET

The datasetw is nothing but the text which the user wants to summarize and visualize better. In our project, we have used a movie plot – “Varisu” and a story plot – “Golden Eggs”. A movie plot consists of many entities so directly reading it and understanding it through text can be challenging. On the other hand, simple everyday stories told to children can also be made to understand better using the graph.

The movie plot consists of 450 words which are summarized into about 150 words. The plot that we got on the internet consisted of many semantic errors which had to be cleaned before we supplied it to the summary generator as it would give a very erroneous graph if supplied without any sort of preprocessing. This manual process ensured that the dataset contained semantically correct sentences which would be very useful for identifying keywords and then building relationships which would be finally translated into graphs. Similarly, the story that we used as our second dataset was also obtained from the internet and had about 300 words and was summarized into 150 words. The same manual cleaning process was again employed to get the required dataset for generating the graph.

Overall, the process of dataset creation is an important and often challenging task in natural language processing. In our project, we had to put some extra effort into cleaning certain aspects of the text so that we had semantically appropriate data which could be converted into graphs.

Chapter 5

RESULTS AND DISCUSSIONS

3.3 Summary of the input text

The project's goal is to make it easier for users to effectively and efficiently visualize their material. By providing a simple-to-use visualization tool, users can quickly comprehend the essence of their text, saving significant time and effort. This allows them to focus on other crucial tasks.

The visualization tool is designed for various applications, including business analysis and academic research. Researchers can use it to visualize large datasets, identifying patterns and connections, while business analysts can analyze customer feedback to uncover insights and opportunities for improvement. The tool features an intuitive and user-friendly interface, making it accessible to users of all technical proficiency levels. By entering their content into the application and selecting the necessary options, users can instantly generate sub-graphs for better understanding.

A key enhancement in this project is the development of an interactive localhost website to host the visualization tool. This website provides a seamless and dynamic user experience, allowing users to interact with their visualized data in real time. The interactive interface enhances usability by enabling smooth navigation, easy customization, and instant modifications to the generated visualizations.

The project was designed to improve productivity and efficiency across multiple industries. By offering a powerful visualization tool, it helps users gain deeper insights into their material and make more informed decisions. With its combination of a user-friendly interface, strong visualization capabilities, and a newly built interactive localhost platform, this project is a valuable resource for anyone needing to analyze complex text data.

5.2 Snapshots

To provide a clearer understanding of the functionality and user experience of the developed visualization tool, this section includes snapshots of the interactive localhost website and the generated relationship graphs.

The localhost website serves as the primary interface where users can input textual data and generate visual representations. The tool features an intuitive design, allowing users to easily interact with and customize their visualizations. The relationship graphs illustrate the connections within the data, helping users identify patterns and insights efficiently.

The following images showcase the main interface of the localhost website, illustrating the process of generating visualizations based on user input. They also include example relationship graphs that demonstrate how the tool structures and displays data connections, providing a clear representation of the summarized information. These visuals provide a practical representation of the tool's capabilities, emphasizing its usability and effectiveness in simplifying complex data analysis.

Fig 5.2.1 shows the snapshot of the input section of the tool.

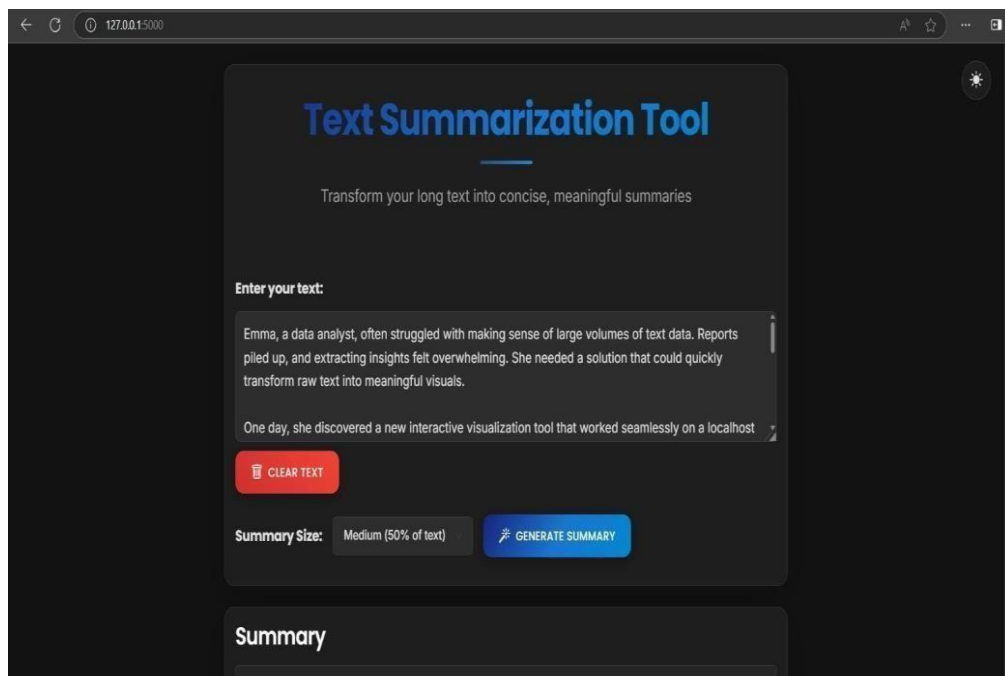


Figure 5.2.1 Snapshot of input section.

Fig 5.2.2 shows the snapshot of the summary section of the tool.

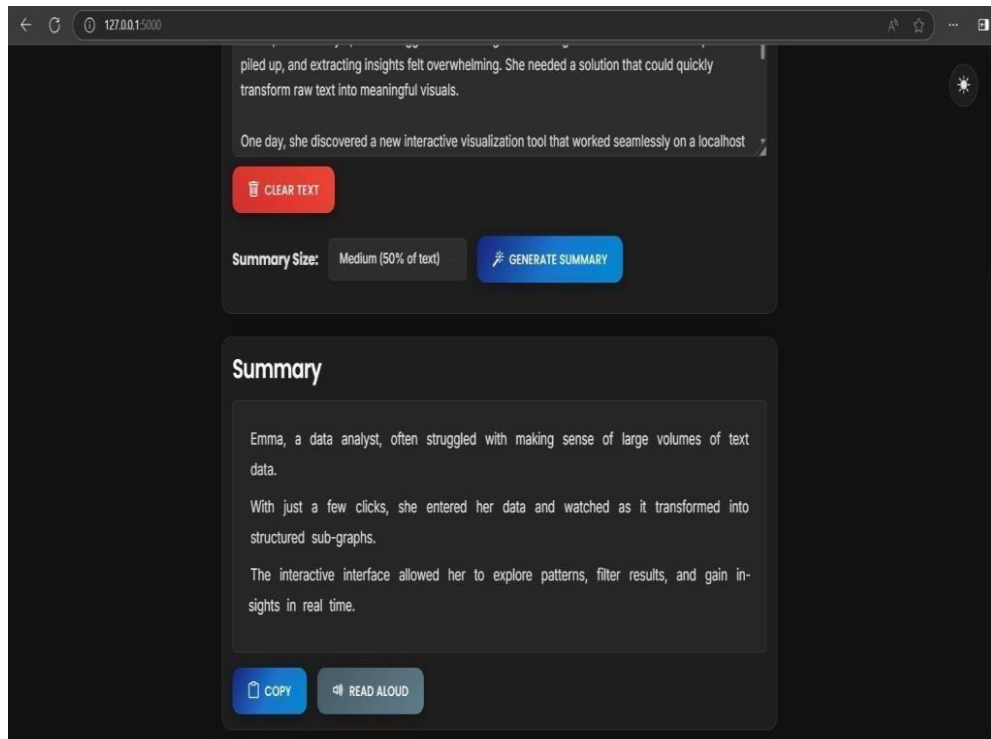


Figure 5.2.2 Snapshot of summary section.

5.3 Relationship.

Fig 5.3.1 shows the snapshot of the relationship obtained from the generated from input text.

```
Entities: [('The Jungle's Secret Deep', 'WORK_OF_ART'), ('Tiko', 'PERSON'), ('Tiko', 'PERSON'), ('One', 'CARDINAL'), ('Azul', 'PERSON'), ('Azul', 'PERSON'), ('Tiko', 'PERSON'), ('Azul', 'PERSON'), ('the Hidden Falls', 'LOC'), ('Tiko', 'PERSON'), ('Azul's', 'PERSON'), ('Tiko', 'PERSON'), ('Tiko', 'PERSON')]
Relationships: [('trees', 'whispered', 'secrets'), ('rivers', 'sang', 'songs'), ('monkey', 'swung', 'vine'), ('monkey', 'swung', 'vine'), ('he', 'noticed', 'parrot'), ('he', 'perched', 'tree'), ('he', 'watching', 'him'), ('I', 'looking', 'something'), ('Azul', 'fluffed', 'feathers'), ('you', 'visit', 'falls'), ('jungle', 'shares', 'secrets'), ('jungle', 'shares', 'those'), ('Tiko', 'followed', 'directions'), ('Tiko', 'navigating', 'vines'), ('Tiko', 'glowing', 'mushrooms'), ('he', 'heard', 'roar'), ('he', 'reached', 'falls'), ('he', 'saw', 'something'), ('he', 'reflecting', 'jungle'), ('Tiko', 'saw', 'reflection'), ('Tiko', 'saw', 'animals'), ('jaguar', 'was', 'him'), ('jungle', 'showing', 'him'), ('jungle', 'showing', 'world'), ('it', 'revealing', 'what'), ('he', 'returned', 'troop'), ('he', 'searched', 'something'), ('he', 'shared', 'wisdom'), ('he', 'teaching', 'others'), ('he', 'appreciate', 'mysteries'), ('he', 'hidden', 'sight'), ('jungle', 'whispered', 'secret'), ('discoveries', 'come', 'those'), ('who', 'seek', 'eyes')]
```

Figure 5.3.1 Snapshot of relationship.

5.4 Knowledge graph

Fig 5.4.1 shows the snapshot of the knowledge graph obtained from the relationships.

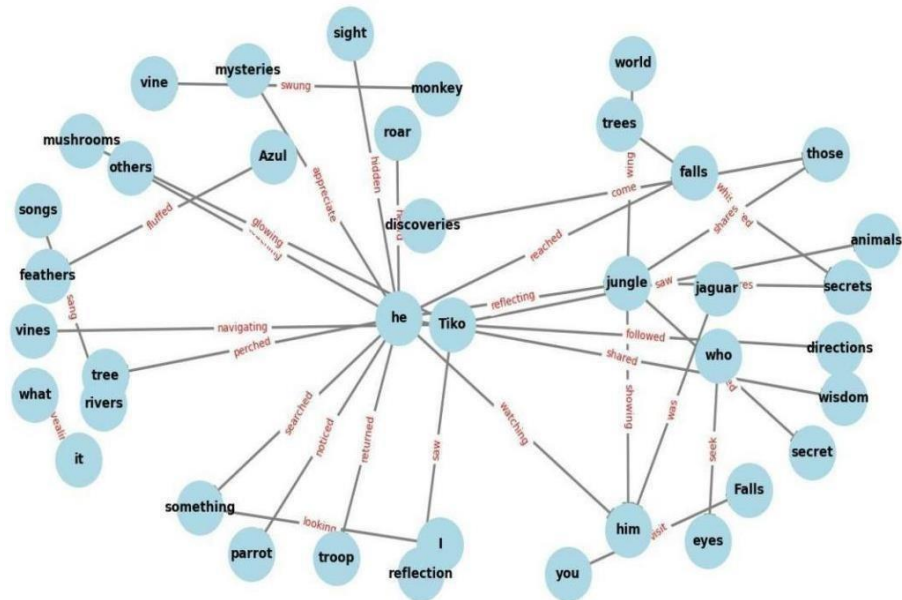


Figure 5.4.1 Snapshot of knowledge graph from generated relationship.

Chapter 6

CONCLUSION

The project began with a literature survey on various text summarization techniques, involving extensive research on existing research papers, articles, and other resources. This step provided a comprehensive understanding of different approaches to text summarization.

Following the literature review, key text summarization algorithms were implemented and evaluated through a comparative study to determine their effectiveness.

In the second phase, a dataset containing movie plot and story plot data was collected. The dataset underwent pre-processing to correct any errors or inconsistencies, ensuring clean and structured data for analysis.

After pre-processing, text summarization techniques were applied to generate concise summaries of the dataset. The next step involved identifying relationships between entities within the text, which were then used to construct a knowledge graph. This graphical representation of entities and their relationships provided a clear and intuitive way to understand the underlying structure of the text.

The developed visualization tool, hosted on an interactive localhost website, enabled users to explore these relationships dynamically, enhancing their ability to comprehend and analyze large volumes of text efficiently.

Chapter 7

SCOPE FOR FUTURE WORK

As the project continues to evolve, several enhancements can be made to improve the accuracy, efficiency, and usability of the knowledge graph. One of the key areas for improvement is the pruning of unnecessary relations that may arise due to inconsistencies in the textual data. Large datasets often contain redundant or irrelevant connections, which can make the knowledge graph cluttered and harder to interpret. By implementing advanced filtering techniques, the system can remove these unnecessary relations, ensuring that only the most meaningful and relevant connections are retained. This will result in a clearer and more insightful visualization of the text data.

Another important aspect to consider is the optimization of text pre-processing techniques. Pre-processing is a crucial step in any natural language processing (NLP) pipeline, as it directly affects the quality of the generated graph. Large text datasets require efficient handling methods to remove noise, correct inconsistencies, and extract key information while minimizing computational overhead. Enhancing tokenization, named entity recognition (NER), and relation extraction techniques can significantly improve the overall performance of the system.

Additionally, integrating machine learning-based approaches for automatic pattern recognition and entity linking could further enhance the accuracy of the knowledge graph. By leveraging AI-driven methods, the system can better identify contextual relationships between different entities, leading to deeper insights from the textual data.

Overall, the future scope of this project focuses on three key areas:

1. Pruning unnecessary relations to improve clarity and reduce redundancy in the graph.
2. Enhancing text pre-processing techniques to make the system more efficient and scalable.
3. Exploring machine learning-based methods for better entity recognition and relationship extraction.

By implementing these improvements, the project can become a powerful and versatile tool for various fields, including business analysis, academic research, and data-driven decision-making.

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